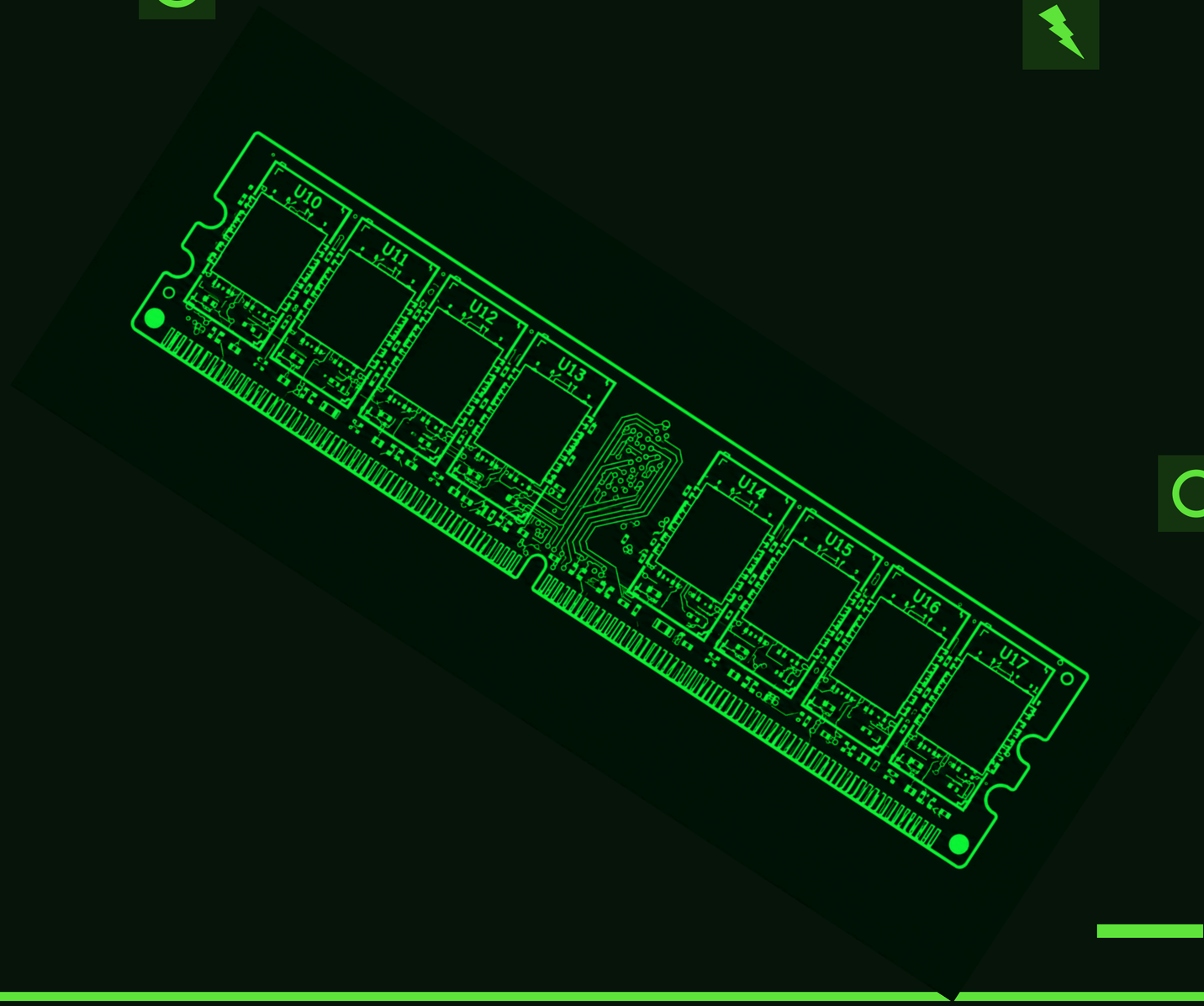
>RAM

Miriam Portales Ortiz



>TABLE OF CONTENTS

01 Introduction

02 Types of memory

03 How RAM works

04 Memory pyramid

05 CPU and motherboard

06 DDR4 vs DDR5

07 DIMM, SO-DIMM and RAM kits

08 Kahoot quiz

>INTRODUCTION

RAM = Random Access Memory

What is RAM?

RAM is the main memory of a computer.

What does it store?

The data and instructions the CPU is using right now: the operating system, open programs and files you edit.

Why is it fast?

It is much faster than a storage drive (HDD or SSD), so the CPU does not wait for the disk.

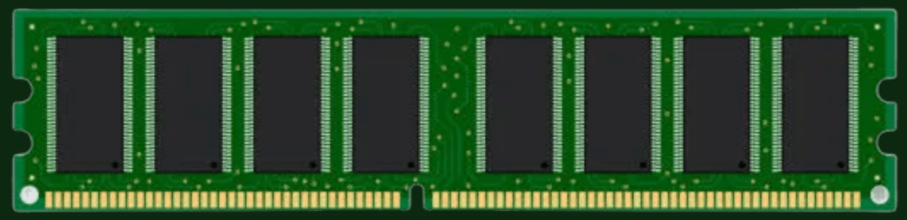
Why more RAM?

The more RAM a computer has, the more programs it can run at the same time.

>VOLATILE VS NON-VOLATILE MEMORY

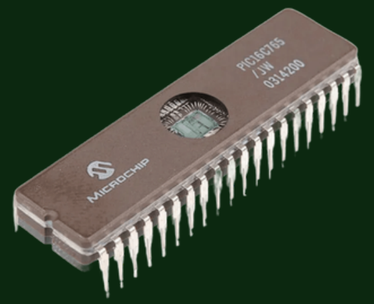
Volatile

Loses its information when the computer turns off.
Stores what the CPU needs quickly while programs are running.
Examples: RAM, cache



Non-volatile

Keeps its information when the computer turns off.
Stores data and instructions for the long term.
Examples: ROM, SSDs



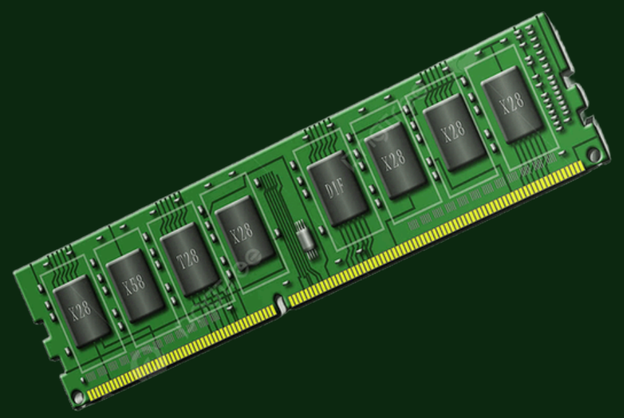


>RAM, CACHE AND ROM

They differ in volatility, speed and function.

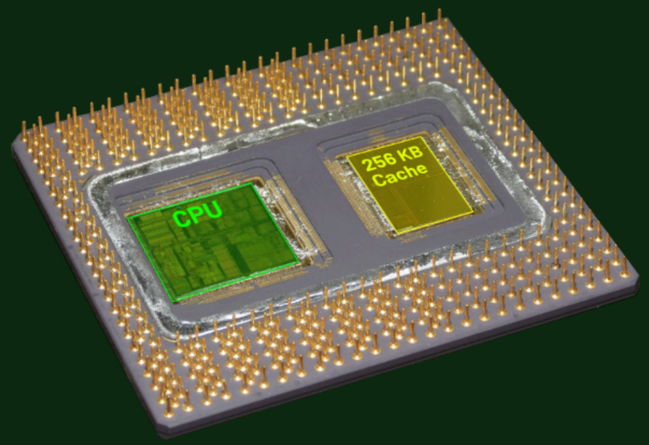
RAM

Volatile. Stores data and instructions for immediate execution. More capacity helps multitasking.



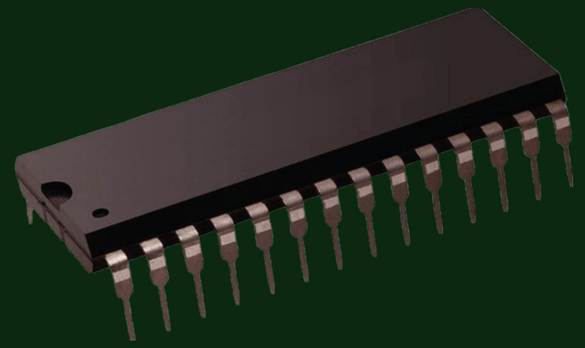
Cache

Volatile. Keeps copies of frequently used RAM data so the CPU works faster. Faster than RAM, but smaller.



ROM

Non-volatile. Its content is not easily changed by the user. Stores start-up instructions and basic hardware settings.



>HOW RAM WORKS

1. Open a program

The operating system copies it from the drive (slow, non-volatile) into RAM (fast, volatile).

2. Work

The CPU reads instructions and data from RAM and writes the results back.

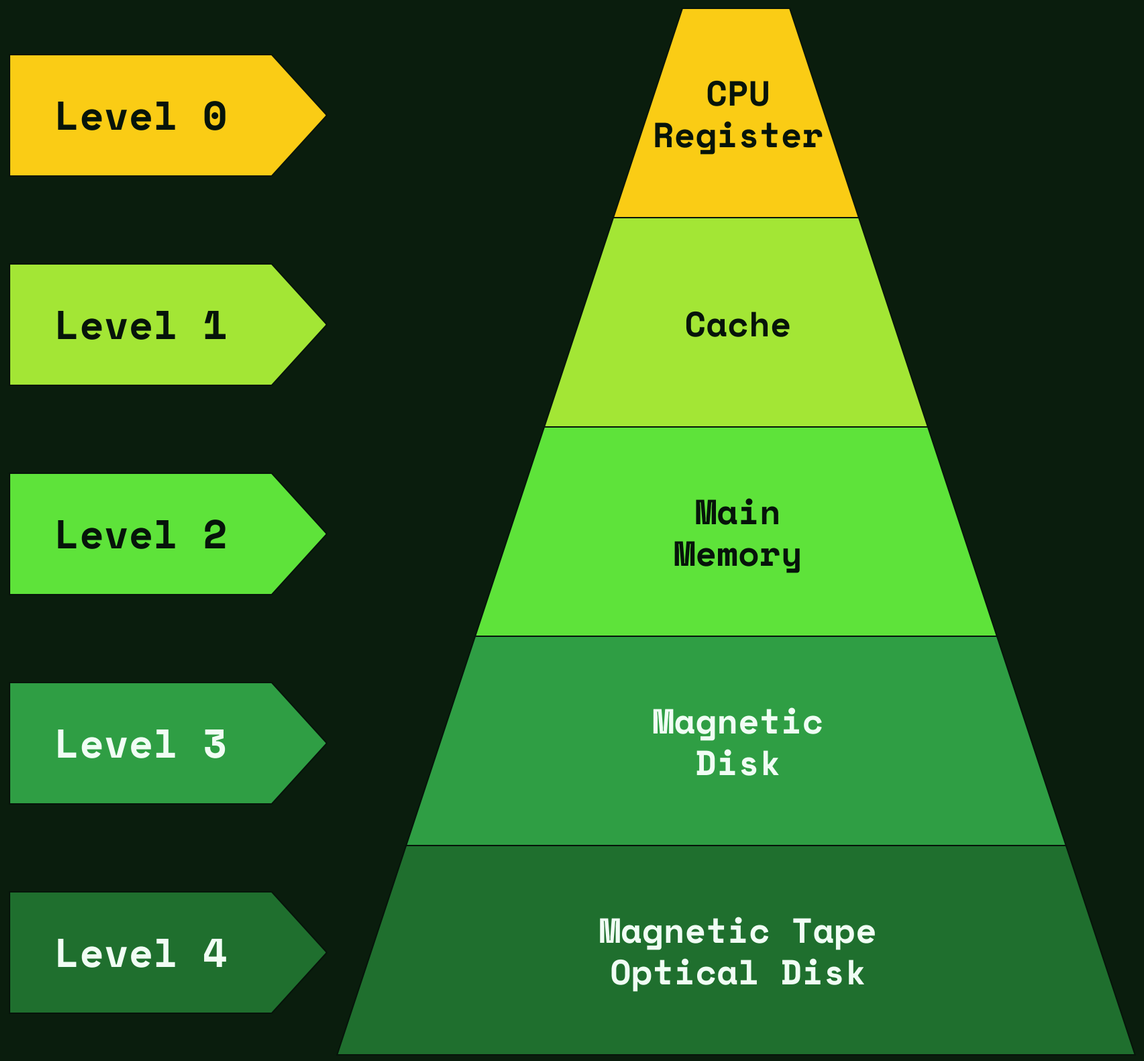
3. Close or turn off

When you close the program or turn off the computer, the RAM is cleared.

4. Not enough RAM?

The system uses part of the drive as virtual memory, and the computer becomes much slower.

>PYRAMID MEMORY



Top: faster, smaller. Bottom: slower, bigger.

Level 0 – CPU Registers
The fastest and smallest memory. Inside the CPU, it holds the data in use right now.

Level 1 – Cache Memory
Very fast memory close to the CPU. Stores frequently used data.

Level 2 – Main Memory (RAM)
Holds the programs and data in use right now. Faster than disks, but loses its data when turned off.

Level 3 – Magnetic Disk
HDDs store files, programs and the operating system. More capacity than RAM, slower, keeps data when turned off.

Level 4 – Magnetic Tape and Optical Disk
Tapes, CDs and DVDs store or back up data. Slow to access, but keep data for a long time.

>CPU AND RAM

The CPU talks to RAM through the memory controller, which is inside the processor. So the CPU decides which memory the computer can use.

Memory type

A CPU can support DDR4, DDR5 or both, depending on the model.

Speed and channels

Each CPU has a maximum memory speed and a number of channels it supports. Faster RAM works at higher speeds only with profiles like XMP or EXPO.

Cache

The CPU keeps copies of the most used RAM data in its cache (L1, L2, L3), so it does not wait for RAM every time.

>MOTHERBOARD AND RAM

RAM modules go in the DIMM slots of the motherboard, so both must be compatible.

DDR generation

A DDR4 module does not fit in a DDR5 slot because the notch is in a different position. Buy RAM of the same generation as the motherboard.

Number of slots

Motherboards usually have 2 or 4 slots. Each one has a maximum total capacity.

Dual channel

The slots work in pairs. For dual channel, install the modules in the correct slots (the manual shows which).

Chipset and BIOS/UEFI

They define the supported speeds. In the BIOS/UEFI you can turn on the XMP or EXPO profile of the RAM.

>DDR4 VS DDR5

DDR4

It is an older type of RAM. It is fast and works well for everyday tasks and games. It uses less power than older memory types.

DDR5

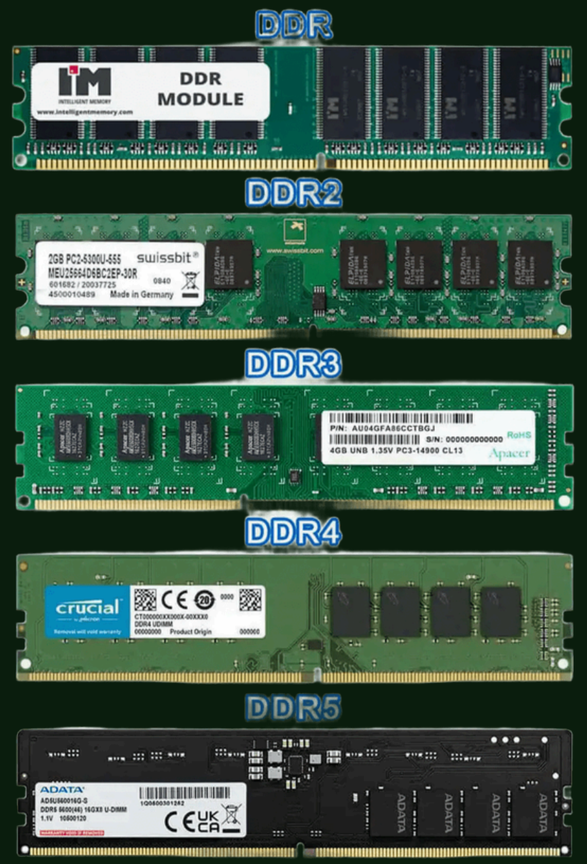
It is a newer type of RAM. It can transfer data faster and offers more memory capacity. It is more energy-efficient than DDR4.

Capacity

They both have different capacities like 8 GB, 16 GB, 32 GB, but DDR5 can support higher capacities per module.

Physical differences

The notch is in a different position on DDR4 and DDR5. This stops you from inserting the wrong type of memory and makes you install it in the correct direction.



>DDR4 VS DDR5 IN DETAIL

MT/s = million transfers per second. A lower CL means a faster response, but speed also matters, so a lower number is not always faster. Dual channel uses two channels instead of one, giving more bandwidth and performance.
CAS latency (CL) =time delay between data read command and availability of data in a computer's RAM

Capacity

Both come in 8, 16 and 32 GB modules, but DDR5 can reach higher capacities per module.

Frequency

Measured in MT/s (shops often write MHz). DDR4 is usually around 3200 MT/s. DDR5 is faster: 6400 MT/s or more.

Voltage

DDR4 works at 1.2 V and DDR5 at 1.1 V, so DDR5 uses less power.

Latency

How long the RAM takes to respond. DDR4: CL16 to CL22. DDR5: CL30 to CL40. DDR5 has a higher CL, but also a higher speed, so the real latency is similar (about 10 ns vs 12.7 ns). Real latency in ns = CL × 2000 / MT/s.

Single vs dual channel

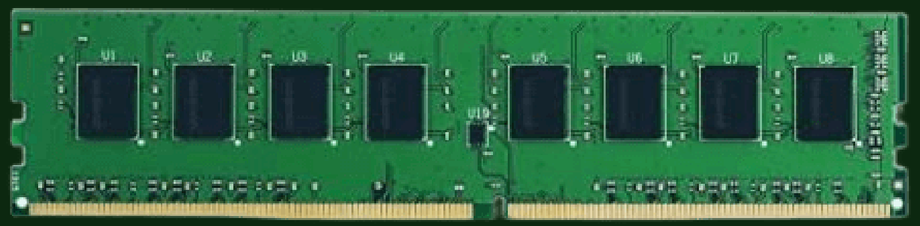
Dual channel uses two channels at the same time, giving more bandwidth and performance. DDR4 has one 64-bit channel per module and DDR5 has two 32-bit subchannels.

>DIMM VS SO-DIMM

DIMM

Bigger. Used in desktop computers, which have more space inside.

DDR4: 288 pins · DDR5: 288 pins



SO-DIMM

Smaller. Used in laptops, which have less space inside.

DDR4: 260 pins · DDR5: 262 pins



They work the same way: both store data and help the computer run programs. The difference is their size, and performance depends on speed and other specifications.

>RAM KITS: DDR4

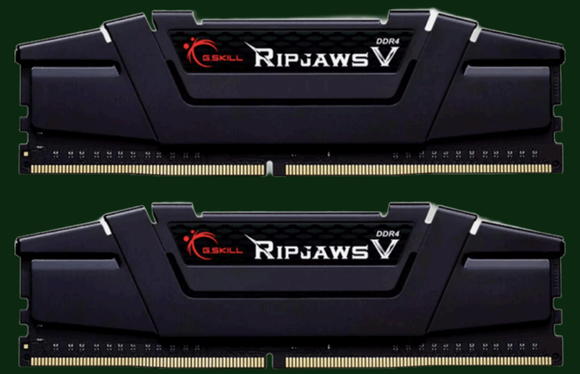
Forgeon Cyclone PLUS V2 DDR4

Capacity: 16 GB (2×8 GB)
Frequency: 3200 MHz
Latency: CL16
149.90€



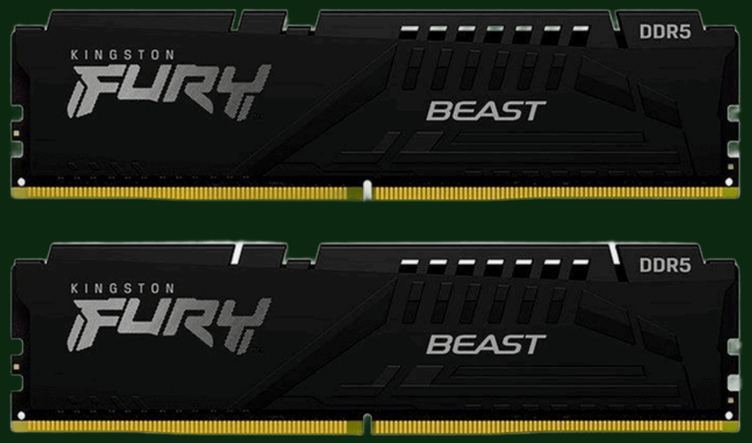
G.Skill Ripjaws V DDR4

Capacity: 32 GB (2×16 GB)
Frequency: 3600 MHz
Latency: CL18
286.99€

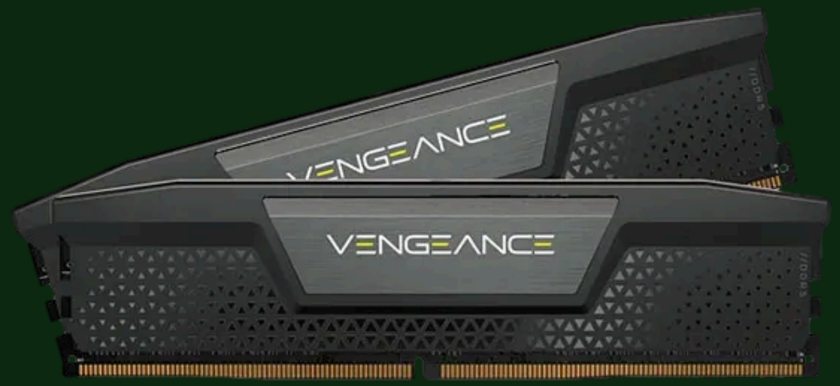


>RAM KITS: DDR5

Kingston FURY Beast DDR5
Capacity: 32 GB (2×16 GB)
Frequency: 5600 MHz
Latency: CL36
649.90€



Corsair Vengeance DDR5
Capacity: 32 GB (2×16 GB)
Frequency: 6000 MHz
Latency: CL38
756.02€



>KAHOOT QUIZ

Test what you learned!

Play my quiz here:

https://create.kahoot.it/share/ram-test/7a3f937f-a1da-4a97-8288-bdef654acedf

>THANKS!

Thank you for your attention